



UNITY V2 INTELLIGENCE ENGINE

PLATFORM ARCHITECTURE & DATA STORAGE BLUEPRINT

CONFIDENTIAL • KINGDOM 3418 HIGH COMMAND

1. EXECUTIVE PLATFORM TOPOLOGY

Unity V2 is an enterprise intelligence operations engine engineered for Rise of Kingdoms. It decouples high-speed public tools from private war-room portals, routing operations through serverless Next.js edge handlers to AWS DynamoDB single-table storage and zero-cost Google Gemini multimodal LLM inference.

END-TO-END OPERATIONAL PIPELINE

1. GOVERNOR CLIENT

Edge React 19 browser interface. HTML5 Canvas WebGPU in-browser optical processing, LocalStorage session caching, and 12-language localization.

React 19 WebGPU / Canvas
12 LocalEs

2. EDGE SERVERLESS

Next.js 16 App Router handlers executing non-blocking telemetry logging, ExcelJS streaming roster ingestion, and multimodal OCR payloads.

Next.js 16 ExcelJS 4.4
NextAuth v5

3. AWS DYNAMODB CORE

High-throughput Single-Table schema (unity-core-production) indexing 139,000+ governors across 163 kingdoms with sub-10ms queries.

Single Table 90d TTL Sub-10ms

2. AWS DYNAMODB SINGLE-TABLE STORAGE SPECIFICATION

All kingdom scans, player credentials, profiles, dossiers, and telemetry write to a single DynamoDB partition topology. This eliminates relational joins and allows instantaneous roster retrieval:

DATA DOMAIN	PARTITION KEY (PK)	SORT KEY (SK)	PAYLOAD & OPERATIONAL LIFECYCLE
Roster Census	SCAN#{kd}#{date}	GOV#{governorId}	Power, Kill Points, T1-T5 kills, Deaths, Tech, Building Power, Alliance Tag. Permanent storage.
Snapshot Index	DATES#{kd}	DATE#{timestamp}	Chronological index of every uploaded scan for that kingdom. Powers timeline scrubbing.
Governor Auth	AUTH_GOVERNOR	GOV#{governorId}	Argon2/SHA256 salted PIN hash, affiliated Kingdom, clearance roles (SuperAdmin / Leader).
Cross-KD Intel	GOV_PROFILE#{id}	PROFILE	Lifetime career trajectory across 163 tracked kingdoms, migration history, gear audit tags.
Officer Dossiers	GOV_NOTE#{id}	NOTE#{timestamp}	Officer intelligence briefs, suspicious behavior flags, rally performance tags, blacklist logs.
Telemetry Log	EVENTS#{YYYY-MM-DD}	EVENT#{ts}#{uuid}	Universal event logging tracking tool engagement. Auto-purged by AWS after 90 days via TTL.
High Command SOS	PENDING_PINGS	PING#{id}	Active officer alert queue dispatched across Discord Webhook, Railway bot, and Admin feed.
Global Config	GLOBAL_CONFIG	CONFIG#{key}	System telemetry calibration cache (139k governors, 163 kingdoms), API models, and keys.

3. ZERO-COST ENGINE MATRIX (\$0.00 SERVERLESS STANDARD)

ExcelJS Streaming Engine

Replaced vulnerable SheetJS (xlsx) with ExcelJS 4.4. Zero memory leaks on 60,000-row KvK workbooks.

\$0.00 / Open Source

Google Gemini Flash-Lite & 2.5

Multimodal OCR & tactical battle reasoning on Google AI Studio Free Tier (20 RPM / 1,500 RPD).

\$0.00 / Free Tier

4. THE COGNITIVE AI ARCHITECTURE (GOOGLE GEMINI)

Unity replaces brittle legacy Python OCR scripts and expensive dedicated GPU servers with serverless multimodal LLMs running at **\$0.00 cost** on Google's generous free tier:

1. VISION OCR SCANNER

gemini-3.1-flash-lite: Scans mobile profile screenshots during autonomous self-registration. Extracts numeric ID, Name, Kingdom, Alliance, Power, and KP in milliseconds.

Instant OCRZero RegexFree Quota

2. AI COMBAT COACH

gemini-3.1-flash-lite / 2.5-pro: 1-on-1 mentor on /stats. Compares player deltas against peer averages. Adapts tone depending on whether the Kingdom is at Peace or War in 12 languages.

Peer DeltaMulti-LingualDynamic Tone

3. TACTICAL BATTLE PREDICTOR

gemini-2.5-flash (Prototype): Head-to-head combat reasoning. Analyzes troop density, T5 fighter percentages, and KvK combat frequency to write ruthless tactical verdicts.

Thinking LLMCombat OddsRadar Metrics

5. SPECIALIZED INTELLIGENCE & WAR ROOM SUITES

MODULE / ROUTE	INTELLIGENCE MODEL	OPERATIONAL PURPOSE IN KINGDOM GOVERNANCE
EK Polygraph /tools/polygraph	Behavioral Delta Matrix	Calculates 30-day ratios between power gain and kill points to unmask farm duelers, spies, and zeroed dead weight.
Player Hunter /tools/hunter	Cross-KD Indexer	Directory across 139,000+ governors in 163 kingdoms. Allows officers to vet migration applicants before accepting passports.
Ghost Hunter /tools/ghost-hunter	Vitality Delta Engine	Detects completely dormant accounts (zero power & KP delta) and calculates Kingdom Vitality Grades (A through F).
Recruitment Hit List /tools/recruitment-hitlist	Velocity Tracker	Tracks departed governors who left your kingdom and are actively accelerating in power elsewhere for re-recruitment.
Anomaly & Fraud Engine /creator/lab/anomaly-detector	Multivariate Outlier (Prototype)	Algorithmic detection flagging stat-padders (>=70% T1 farm kills), scripted bot accounts, and account-buyer liabilities.
AI Vision Scanner /experimental/ocr	WebGPU / Canvas (Prototype)	Dual-engine OCR running optical feature extraction directly on the client GPU with zero API tokens consumed.

6. FAIL-SAFE ALERT & HIGH COMMAND SECURITY PIPELINE

When players experience registration hurdles, system anomalies occur, or emergency support is requested, Unity's multi-channel fail-safe dispatcher fires in parallel:

DISPATCHER PIPELINE: `notifyAdmin({ type: "FAILED_REGISTRATION", ... })`

- Discord Webhook:** Real-time high-priority embed posted to leadership war channel.
- Railway Bot Bridge:** Automated ping dispatched via `/api/system/broadcast`.
- DynamoDB Presence Queue:** Logged to PENDING_PINGS partition for in-dashboard badge indicator.
- Admin Telemetry Feed:** Instant reviewable event card on `/admin/analytics`.